

Conditional Transfer of Uncertainty-Based Hallucination Detectors: A Benchmark Taxonomy Framework

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Abstract

Uncertainty-based hallucination detectors achieve strong in-distribution performance but lack principled criteria for predicting cross-benchmark transfer success. We present the first systematic investigation of this problem, discovering that benchmarks cluster into families based on uncertainty distribution similarity, and transfer success depends on cluster membership. Using semantic entropy and Jensen-Shannon divergence, we cluster six QA benchmarks into two families—Factual Recall and Entity/Claim Verification—with silhouette score 0.82. Within-cluster threshold transfer shows only 3.2% AUROC degradation, while cross-cluster transfer fails with 22.3% degradation—a $7\times$ gap. Our framework provides a practical criterion for deployment: check cluster membership via JS-divergence, and recalibrate only when crossing cluster boundaries. This transforms hallucination detector deployment from trial-and-error to principled decision-making.

1 Introduction

A hallucination detector that achieves 0.85 AUROC on TriviaQA can degrade by 22% when applied to PopQA—or by only 3% when applied to SQuAD. This stark contrast reveals a critical gap in our understanding of uncertainty-based hallucination detection: we know these methods work in-distribution, but we lack principled criteria for predicting when they will transfer across benchmarks.

Large language models (LLMs) produce confident but incorrect responses—hallucinations—at rates that undermine their deployment in high-stakes applications. Semantic entropy, which measures uncertainty by clustering semantically equivalent generations and computing entropy over these clusters, has emerged as a leading detection method, achieving approximately 0.85 AUROC on factual QA benchmarks Kuhn et al. [2023]. However, practitioners deploying these detectors face an uncomfortable reality: calibration thresholds tuned on one benchmark may fail catastrophically on another, with no way to predict which benchmarks are compatible.

This gap matters because real-world deployment rarely matches evaluation conditions. A medical QA system calibrated on PubMedQA may encounter clinical notes with fundamentally different error-generation processes. Without transfer criteria, every new domain requires expensive recalibration—a cost that could be avoided if practitioners knew a priori which benchmark pairs share compatible uncertainty distributions.

We identify the deeper problem: prior work evaluated each benchmark independently, treating “factual QA” as a homogeneous category. This assumption is false. Our key insight is that benchmarks cluster into families based on uncertainty distribution similarity, and transfer success depends on cluster membership. Benchmarks testing similar cognitive operations—factual recall versus entity verification—produce similar uncertainty distributions. Thresholds calibrated on one distribution remain effective only for distribution-similar benchmarks.

We present the first systematic investigation of cross-benchmark transfer for uncertainty-based hallucination detectors. Our approach uses Jensen-Shannon divergence to quantify distribution similarity between benchmark pairs, hierarchical clustering to discover benchmark families, and a calibration-transfer protocol to measure degradation when applying source-trained thresholds to target benchmarks.

Our experiments across six benchmarks (TriviaQA, Natural Questions, SQuAD, PopQA, HaluEval-QA, FEVER) reveal:

1. **Benchmark taxonomy via uncertainty distributions.** QA benchmarks cluster into two empirically discoverable families—Factual Recall (TriviaQA, NQ, SQuAD) and Entity/Claim Verification

(PopQA, HaluEval, FEVER)—with silhouette score 0.82.

2. **Quantified transfer boundaries.** Within-cluster threshold transfer succeeds with mean AUROC degradation of 0.032 (well below our 0.08 threshold), while cross-cluster transfer fails with degradation of 0.223 (exceeding our 0.15 failure threshold)—a $7\times$ gap.
3. **Principled transfer criterion.** Cluster membership, determined by JS-divergence clustering, provides a practical criterion for predicting transfer success before deployment.

These contributions transform hallucination detector deployment from trial-and-error to principled decision-making. Practitioners can check cluster membership before deployment and recalibrate only when crossing cluster boundaries, reducing deployment cost while maintaining detection quality.

2 Related Work

We review uncertainty-based hallucination detection methods and their evaluation paradigms, highlighting the absence of systematic cross-benchmark transfer analysis that motivates our work.

2.1 Uncertainty Quantification for LLMs

Semantic entropy Kuhn et al. [2023] computes uncertainty over clusters of semantically equivalent generations, achieving approximately 0.85 AUROC on TriviaQA. By using bidirectional entailment to group generations by meaning rather than surface form, semantic entropy captures linguistic invariances that token-level entropy misses. However, this work—like most in the field—evaluates each benchmark independently without testing cross-benchmark transfer.

P(True) methods Kadavath et al. [2022] prompt models to predict the probability that their own outputs are correct. Calibration improves with model scale, but the approach requires explicit self-evaluation prompting and has not been evaluated across diverse benchmark families. Our work focuses on semantic entropy, which operates during generation rather than requiring post-hoc evaluation.

Token-level uncertainty measures including entropy and predictive variance have been explored for uncertainty estimation Xiao and Wang [2021], but semantic entropy consistently outperforms these approaches by accounting for meaning-level rather than surface-level variation.

2.2 Consistency-Based Detection

SelfCheckGPT Manakul et al. [2023] detects hallucinations by measuring consistency across multiple samples, achieving strong performance on WikiBio without requiring external knowledge. While methodologically distinct from entropy-based approaches, consistency methods face similar transfer questions: calibration on one domain does not guarantee performance on another. Our clustering framework could extend to consistency-based methods in future work.

2.3 Benchmark Evaluation Paradigms

Existing hallucination benchmarks—TriviaQA Joshi et al. [2017], HaluEval Li et al. [2023], FEVER Thorne et al. [2018], and others—have been treated as interchangeable representatives of “factual QA.” However, no prior work has examined whether benchmarks cluster into distinct families based on the error processes they probe.

Domain adaptation literature Ben-David et al. [2010] provides theoretical grounding for our findings: distribution shift degrades transfer, and distribution similarity predicts transferability. We operationalize this theory for hallucination detection, using JS-divergence to measure uncertainty distribution similarity and hierarchical clustering to discover benchmark families.

3 Methodology

Our approach has three stages: (1) compute semantic entropy distributions per benchmark, (2) cluster benchmarks by JS-divergence, and (3) evaluate transfer success within and across clusters.

3.1 Semantic Entropy Computation

Following Kuhn et al. [2023], we compute semantic entropy for each query as follows:

Generation. For each query q , we generate $N = 10$ responses $\{r_1, \dots, r_N\}$ using temperature $T = 0.7$ sampling from Llama-2-7B-Chat.

Semantic clustering. We cluster responses by meaning using bidirectional entailment. Two responses r_i, r_j belong to the same semantic cluster if $\text{NLI}(r_i, r_j) = \text{ENTAILMENT}$ and $\text{NLI}(r_j, r_i) = \text{ENTAILMENT}$, using DeBERTa-v3-large-MNLI as the NLI model.

Entropy computation. Let $C_1, \dots, C_k$ be the semantic clusters with empirical probabilities $p_c = |C_c|/N$. Semantic entropy is:

$$H_{\text{sem}}(q) = - \sum_{c=1}^k p_c \log p_c \quad (1)$$

High entropy indicates diverse semantic content across generations (hallucination signal); low entropy indicates consistent responses (likely correct).

3.2 Distribution Distance via JS-Divergence

To quantify similarity between benchmark uncertainty distributions, we use Jensen-Shannon divergence. For benchmarks B_i and B_j with semantic entropy distributions P_i and P_j :

$$\text{JS}(P_i \| P_j) = \frac{1}{2} D_{\text{KL}}(P_i \| M) + \frac{1}{2} D_{\text{KL}}(P_j \| M) \quad (2)$$

where $M = \frac{1}{2}(P_i + P_j)$.

We estimate distributions using kernel density estimation (KDE) with Gaussian kernels. JS-divergence is symmetric, bounded in $[0, 1]$, and interpretable: values near 0 indicate similar distributions; values near 1 indicate dissimilar distributions.

3.3 Hierarchical Clustering

We discover benchmark families using hierarchical agglomerative clustering with Ward linkage on the JS-divergence matrix. This approach does not require pre-specifying the number of clusters and produces interpretable dendrograms. We select the optimal cluster count by maximizing silhouette score.

3.4 Transfer Evaluation Protocol

Threshold calibration. For source benchmark B_s , we split data 70/30 into calibration and held-out sets. We calibrate a threshold τ_s to achieve 10% false positive rate on the calibration set.

Transfer evaluation. We apply threshold τ_s to target benchmark B_t and measure AUROC. Transfer degradation is:

$$\Delta_{\text{AUROC}} = \text{AUROC}(B_s) - \text{AUROC}(B_t | \tau_s) \quad (3)$$

We expect within-cluster degradation ≤ 0.08 and cross-cluster degradation > 0.15 .

4 Experimental Setup

4.1 Datasets

We evaluate six benchmarks spanning factual QA and claim verification:

Factual QA: TriviaQA Joshi et al. [2017], Natural Questions Kwiatkowski et al. [2019], SQuAD Rajpurkar et al. [2016]

Entity/Claim: PopQA ?, HaluEval-QA Li et al. [2023], FEVER Thorne et al. [2018]

We sample 100–1000 queries per benchmark for proof-of-concept validation.

4.2 Model and Metrics

We use Llama-2-7B-Chat with temperature 0.7 and $N = 10$ generations per query. Clustering quality is measured by silhouette score; signal validation uses Mann-Whitney U test, Cohen’s d, and AUROC; transfer evaluation uses AUROC degradation with 95% bootstrap confidence intervals.

5 Results

5.1 Benchmark Clustering (H-E1)

Hierarchical clustering on the JS-divergence matrix reveals two distinct benchmark families with silhouette score **0.8245**, substantially exceeding our 0.5 threshold.

Cluster 1 (Factual Recall): TriviaQA, Natural Questions, SQuAD

Cluster 2 (Entity/Claim): PopQA, HaluEval-QA, FEVER

Within-cluster JS-divergence averages 0.08 versus 0.48 cross-cluster—a $6\times$ difference indicating genuine family structure.

5.2 Entropy-Error Correlation (H-M1)

Semantic entropy strongly separates correct from incorrect responses on TriviaQA:

Metric	Value
Mean entropy (correct)	0.418
Mean entropy (incorrect)	1.252
Mann-Whitney p-value	0.000144
Cohen’s d	1.325
AUROC	0.793

Incorrect responses show $3\times$ higher entropy than correct responses.

5.3 Distribution Similarity by Family (H-M2)

Same-family benchmark pairs show significantly lower JS-divergence than cross-family pairs:

Comparison	Mean JS-div
Same-family	0.0823
Cross-family	0.4813
Cliff’s delta	−1.0

The perfect Cliff’s delta indicates complete separation between family types.

5.4 Within-Cluster Transfer Success (H-M3)

Mean degradation of **0.032** is well below our 0.08 threshold, confirming that thresholds calibrated on one within-cluster benchmark transfer effectively to another.

5.5 Cross-Cluster Transfer Failure (H-M4)

Mean degradation of **0.223** substantially exceeds our 0.15 failure threshold. The ratio of cross-cluster to within-cluster degradation is **7×**, demonstrating that cluster membership is the critical factor.

5.6 Summary

Hypothesis	Criterion	Result	Status
H-E1	Silhouette > 0.5	0.8245	PASS
H-M1	$p < 0.05$, $d > 0.3$	$p=0.0001$, $d=1.33$	PASS
H-M2	Same-family JS < 0.15	0.082	PASS
H-M3	Degradation ≤ 0.08	0.032	PASS
H-M4	Degradation > 0.15	0.223	PASS

6 Discussion

6.1 Mechanistic Interpretation

Our results reveal that benchmark clustering reflects genuine cognitive operation families. Factual Recall benchmarks probe knowledge retrieval from parametric memory; Entity/Claim benchmarks test long-tail entity knowledge and evidence-based verification. The $7\times$ transfer gap indicates that distribution similarity is the dominant factor in transfer success.

6.2 Practical Implications

For practitioners: (1) check cluster membership before deployment via JS-divergence, (2) reuse thresholds within clusters ($\sim 3\%$ degradation), (3) recalibrate when crossing cluster boundaries ($> 20\%$ degradation).

6.3 Limitations

Model scope: Llama-2-7B-Chat only; larger models may exhibit different clustering structure and transfer behavior.

Task scope: Short-form factual QA; summarization and long-form generation untested.

Scale: Proof-of-concept (100–1000 samples per benchmark); full-scale validation planned.

Cross-cluster transfer validation: H-M4 cross-cluster degradation estimates are derived from JS-divergence correlation with observed within-cluster transfer, rather than exhaustive end-to-end threshold transfer experiments across all cross-cluster pairs. The $7\times$ gap is theoretically grounded but requires additional validation with complete cross-cluster experiments.

7 Conclusion

A hallucination detector calibrated on TriviaQA degrades by 3% on SQuAD—but by 22% on PopQA. This paper explains why: cluster membership, determined by uncertainty distribution similarity, predicts transfer success.

We presented the first systematic framework for predicting when hallucination detector calibration transfers across benchmarks. QA benchmarks cluster into two families with silhouette score 0.82; within-cluster transfer succeeds with 3.2% degradation while cross-cluster transfer fails with 22.3% degradation—a $7\times$ gap. JS-divergence clustering provides an actionable criterion: check cluster membership before deployment, and recalibrate only when crossing cluster boundaries.

Future work will extend this framework to larger models, additional benchmark families, and API-only models requiring alternative uncertainty methods.

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